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Feature Dimensionality Reduction

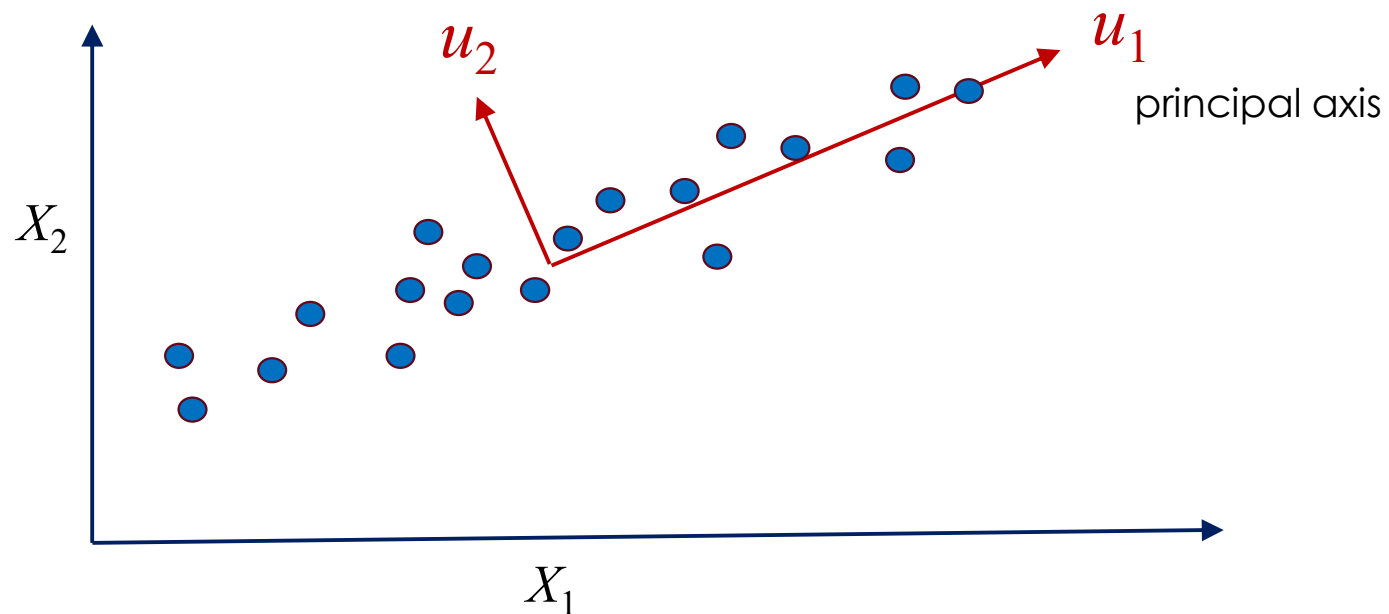
01211373 Machine
Learning and
Programming for
Industry



OUTLINE

- Unsupervised
 - Principal Component Analysis (PCA)
- Supervised
 - Linear Discriminant Analysis (LDA)

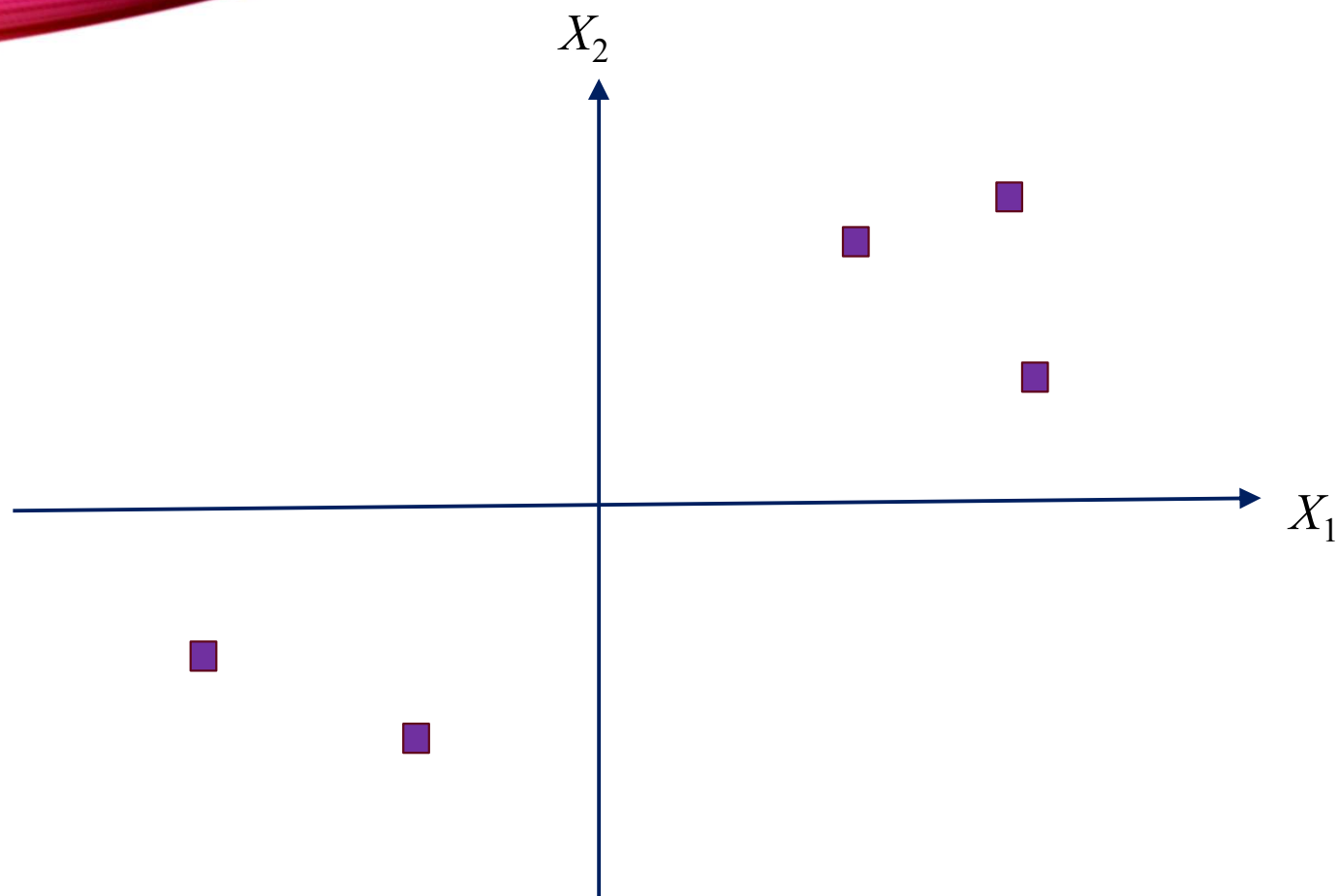
PCA



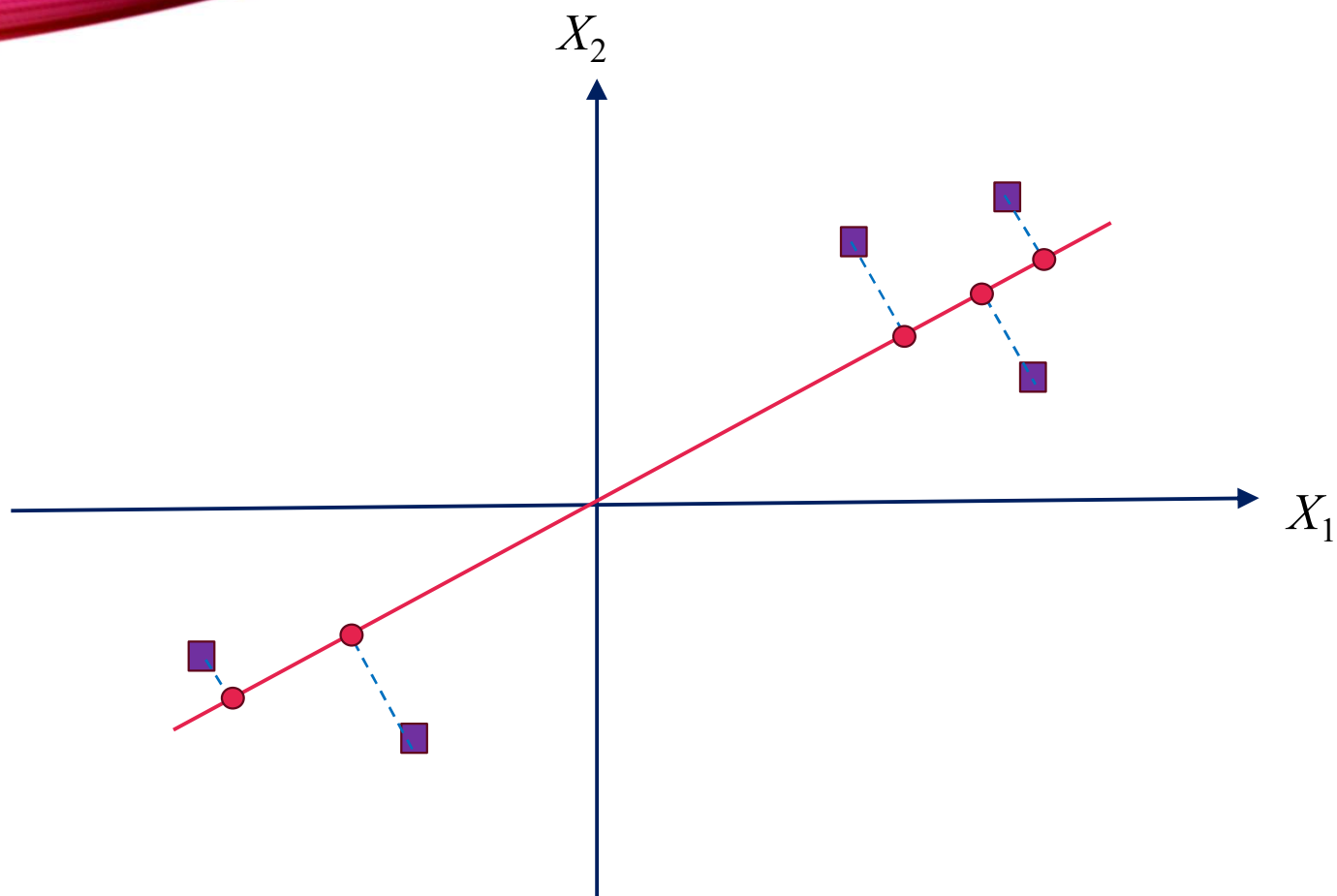
We would like to choose a direction u such that if we were to approximate the data as lying in the direction/subspace corresponding to u , as much as possible of this variance is still retained



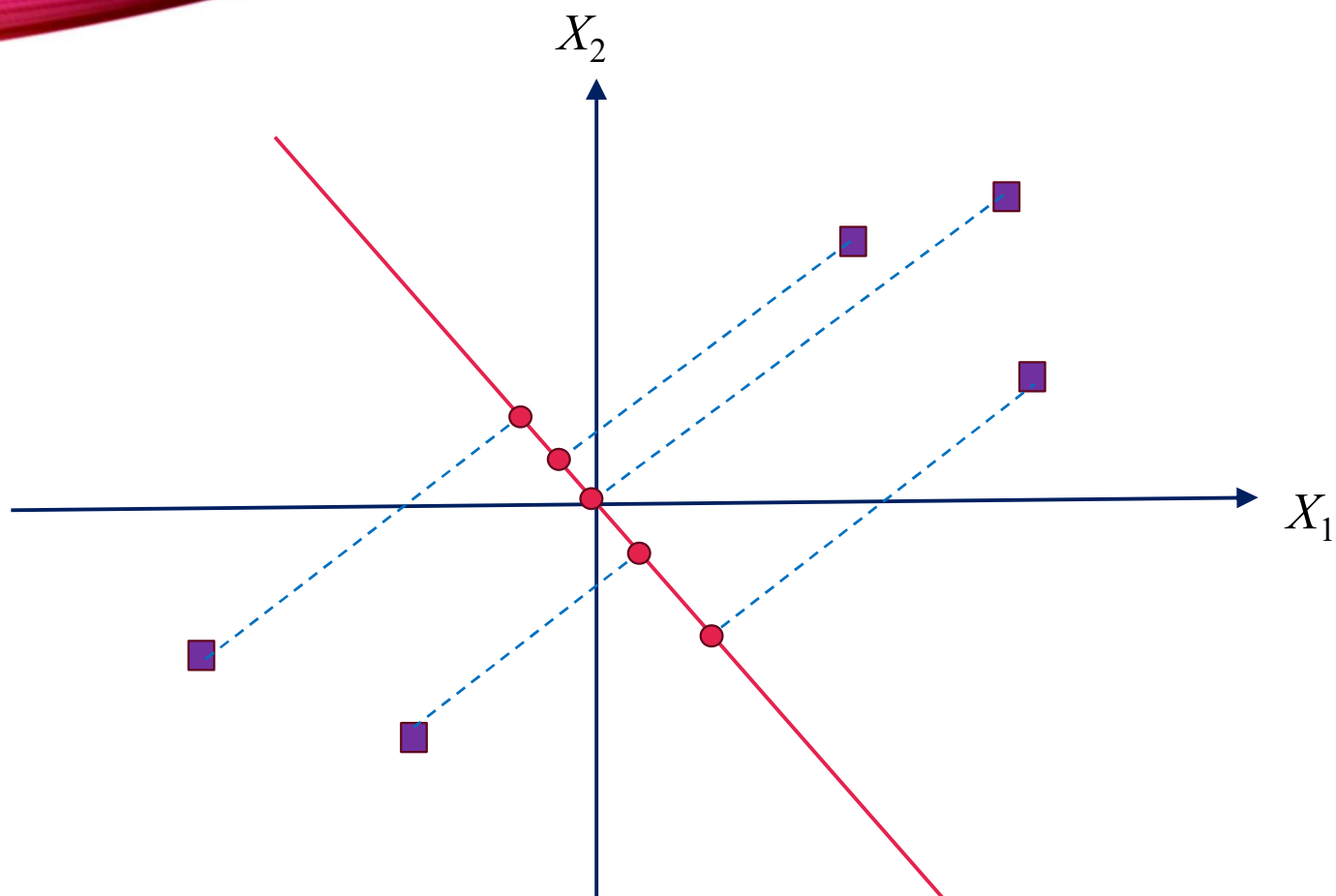
PCA



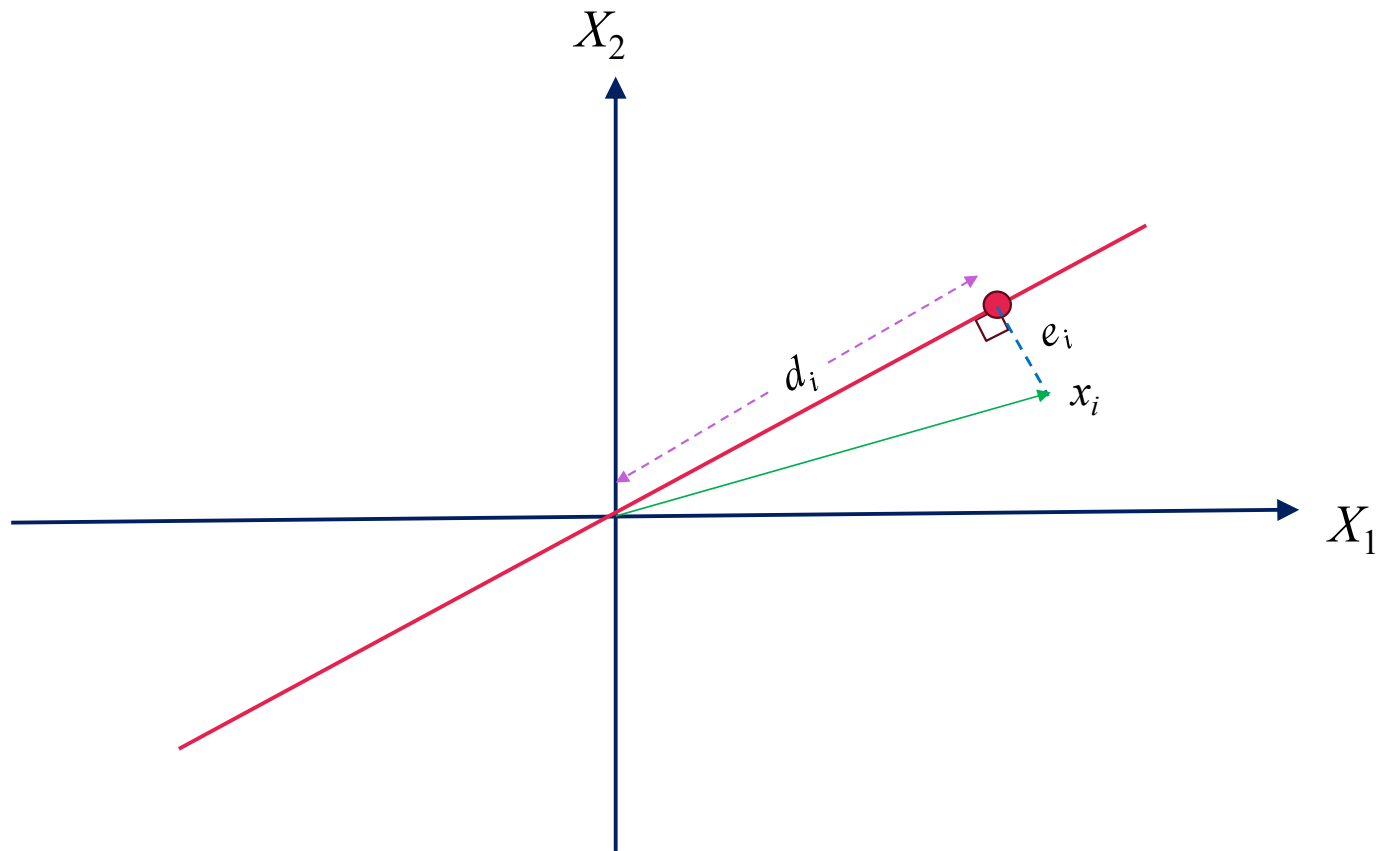
PCA



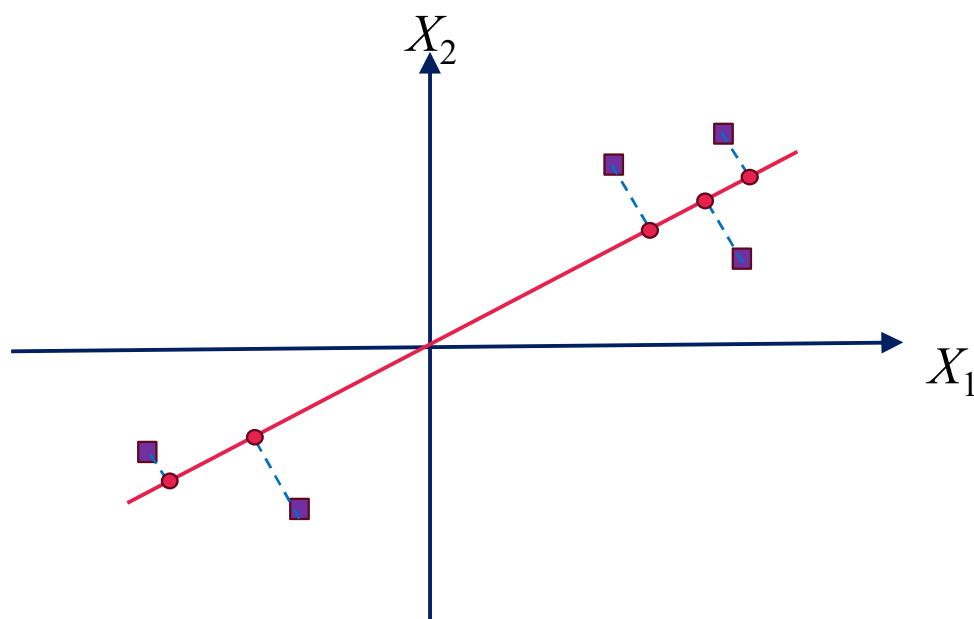
PCA



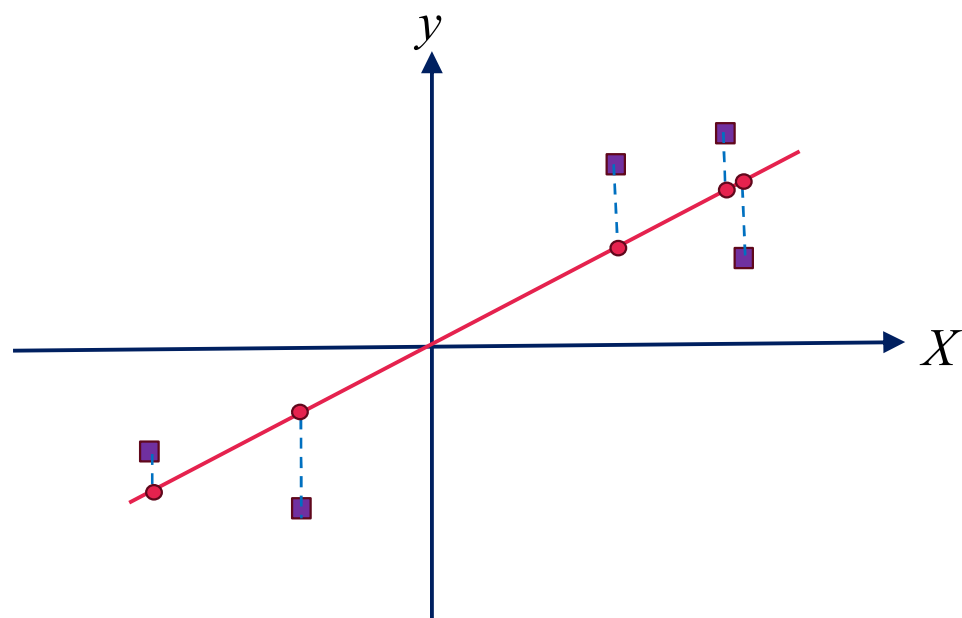
Minimizing error $\longleftrightarrow$ maximizing variance



PCA

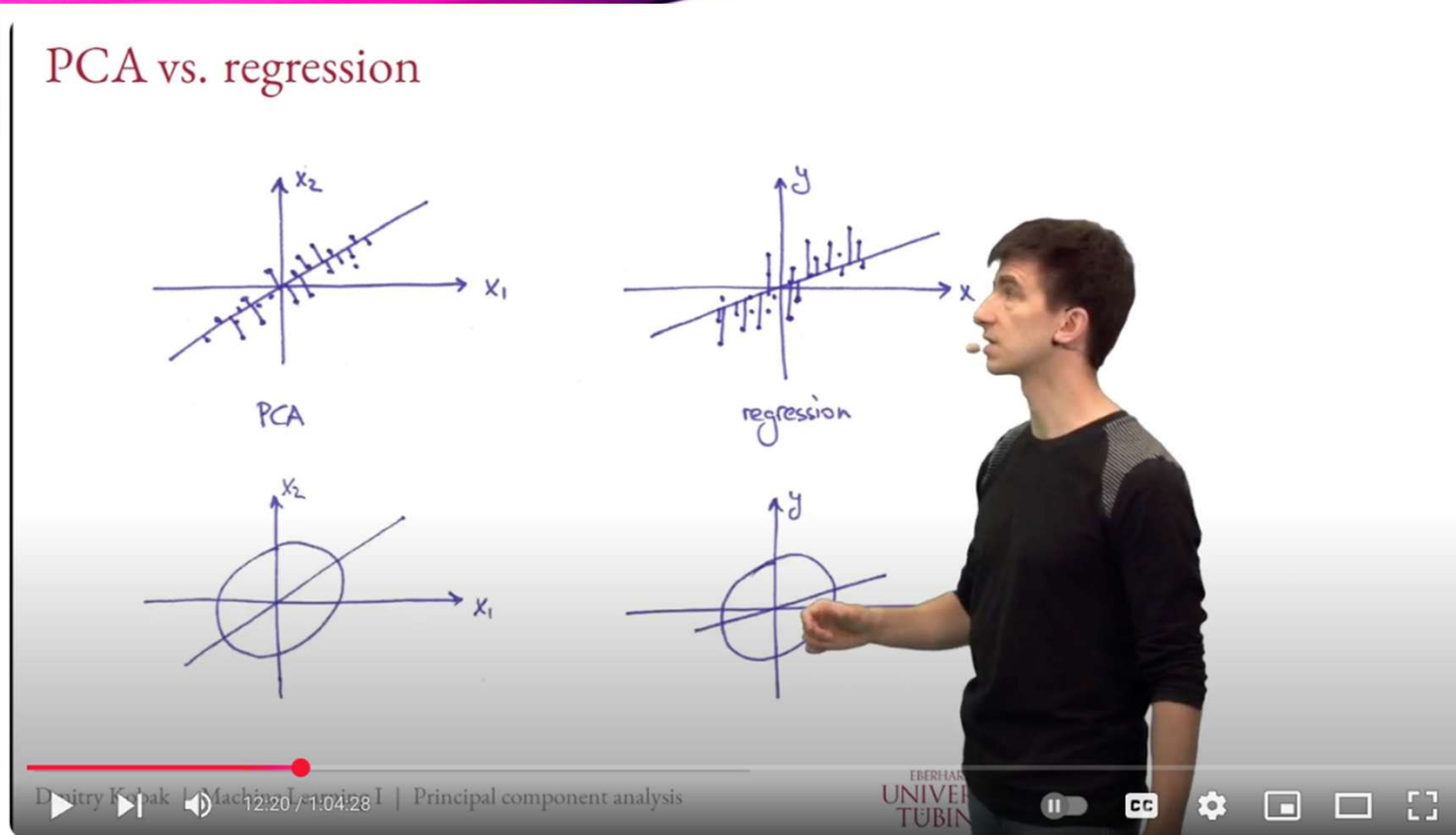


Linear regression



PCA v.s. regression

(watch this video)



Introduction to Machine Learning - 10 - Principal component analysis



Tübingen Machine Learning
46.6K subscribers



166

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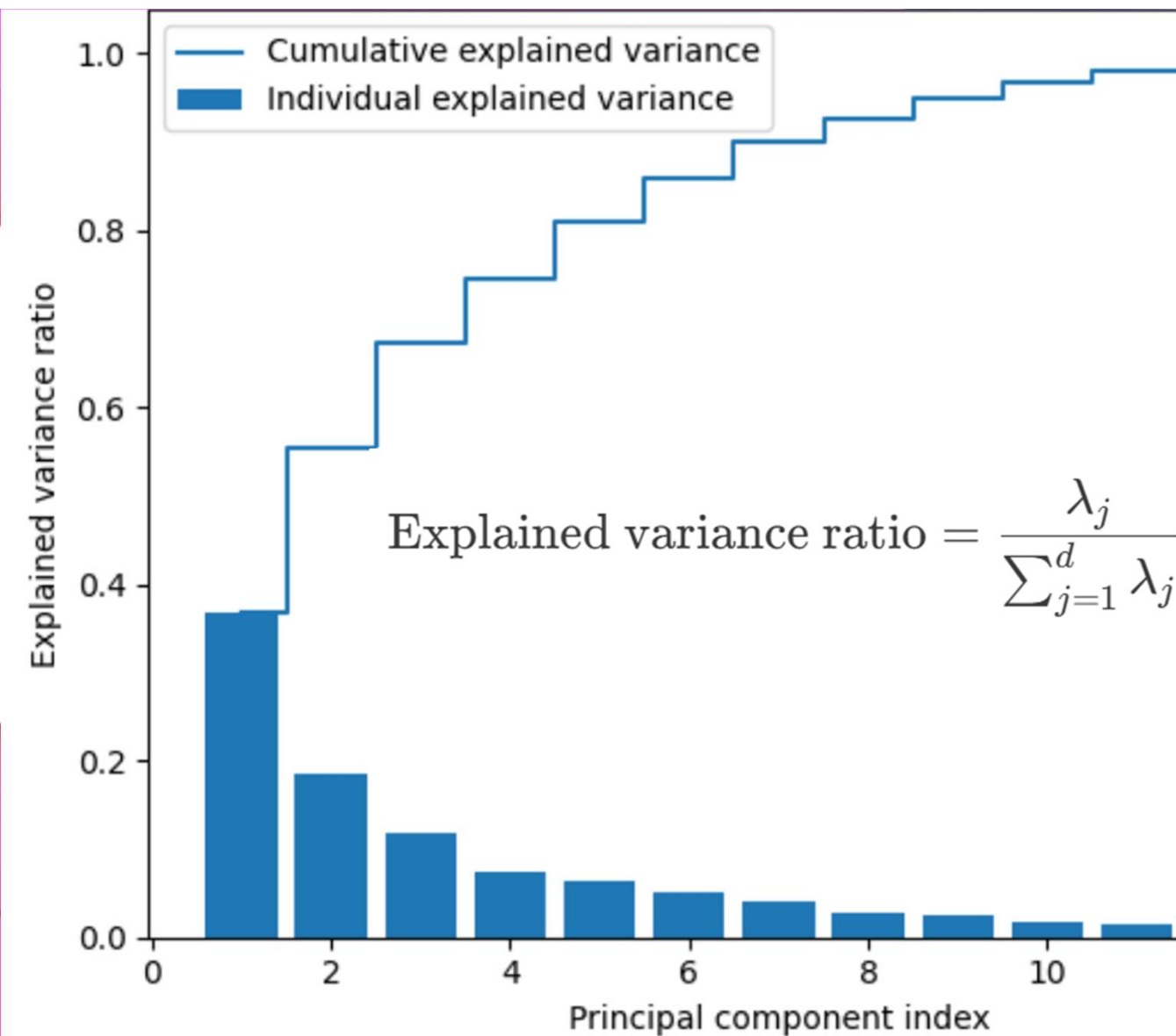
Share



https://youtu.be/xBf_LZ5ZgY4?si=PbUs_KjGuLo9BZfC&t=625

PCA with wine dataset

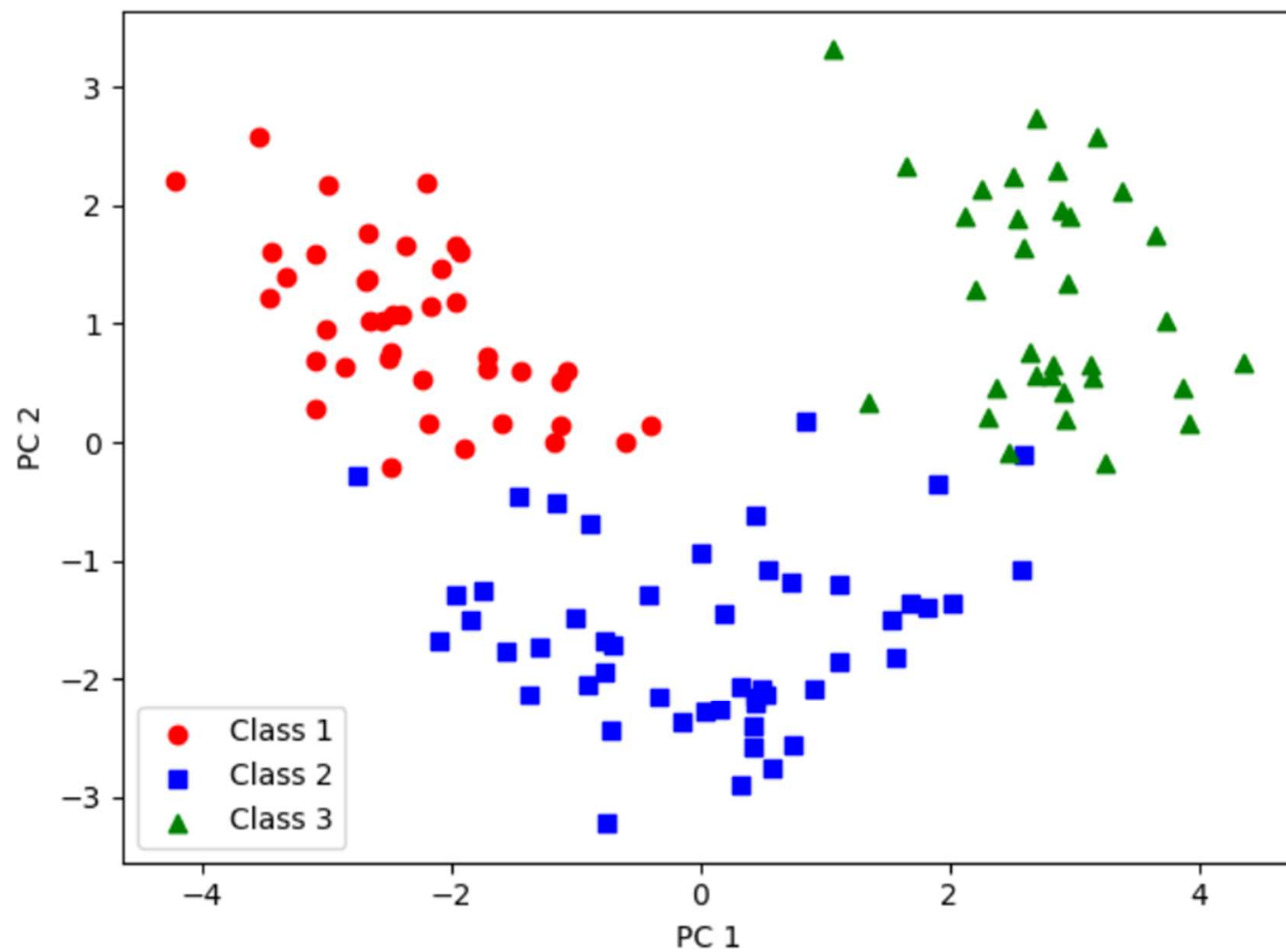
	Class label	Alcohol	Malic acid	Ash	Alcalinity of ash	Magnesium	Total phenols	Flavanoids	Nonflavanoid phenols	Proanthocyanins	Color intensity	Hue	OD280/OD315 of diluted wines	Proline
0	1	14.23	1.71	2.43	15.6	127	2.80	3.06	0.28	2.29	5.64	1.04	3.92	1065
1	1	13.20	1.78	2.14	11.2	100	2.65	2.76	0.26	1.28	4.38	1.05	3.40	1050
2	1	13.16	2.36	2.67	18.6	101	2.80	3.24	0.30	2.81	5.68	1.03	3.17	1185
3	1	14.37	1.95	2.50	16.8	113	3.85	3.49	0.24	2.18	7.80	0.86	3.45	1480
4	1	13.24	2.59	2.87	21.0	118	2.80	2.69	0.39	1.82	4.32	1.04	2.93	735



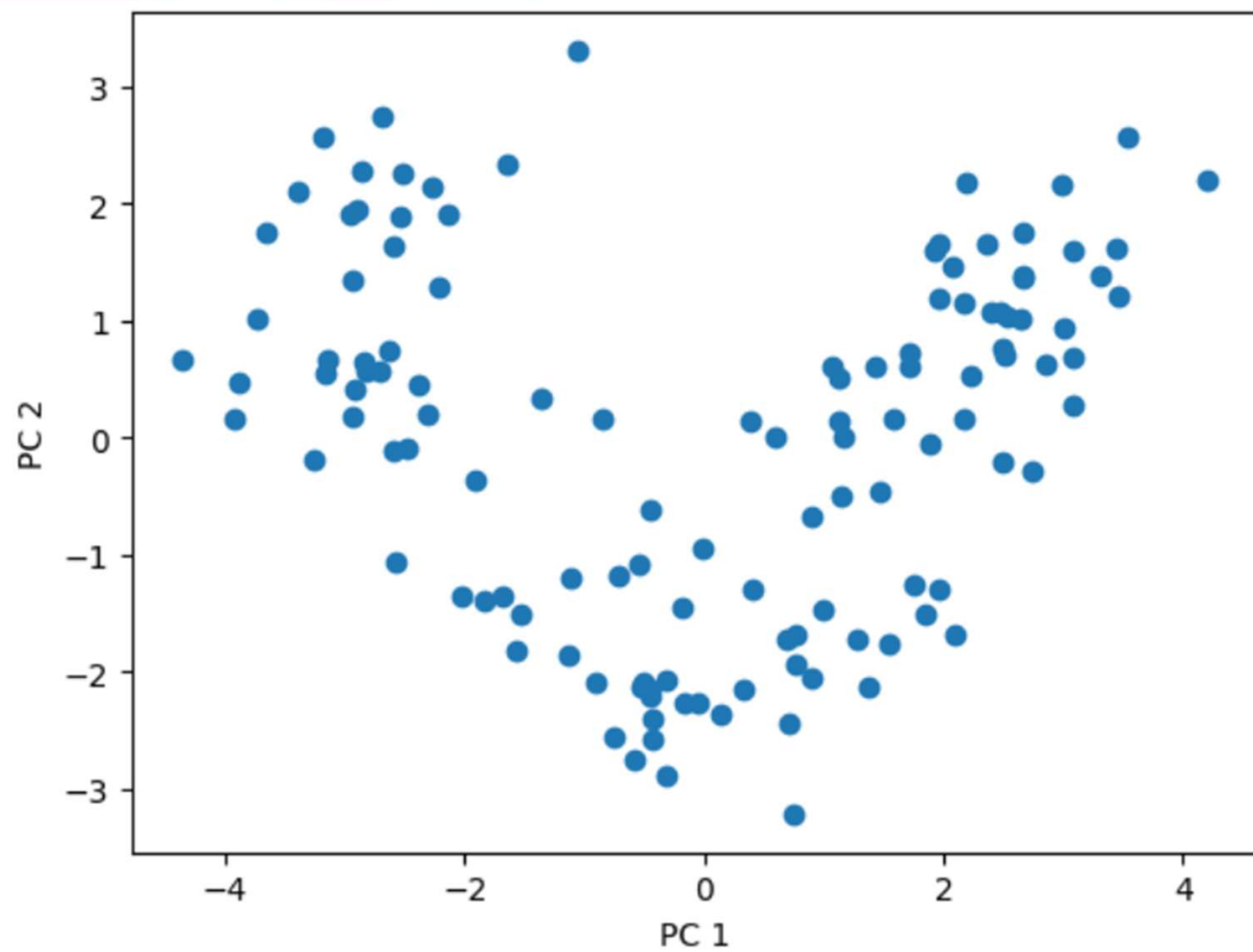
$$\text{Explained variance ratio} = \frac{\lambda_j}{\sum_{j=1}^d \lambda_j}$$

Total and
explained
variance



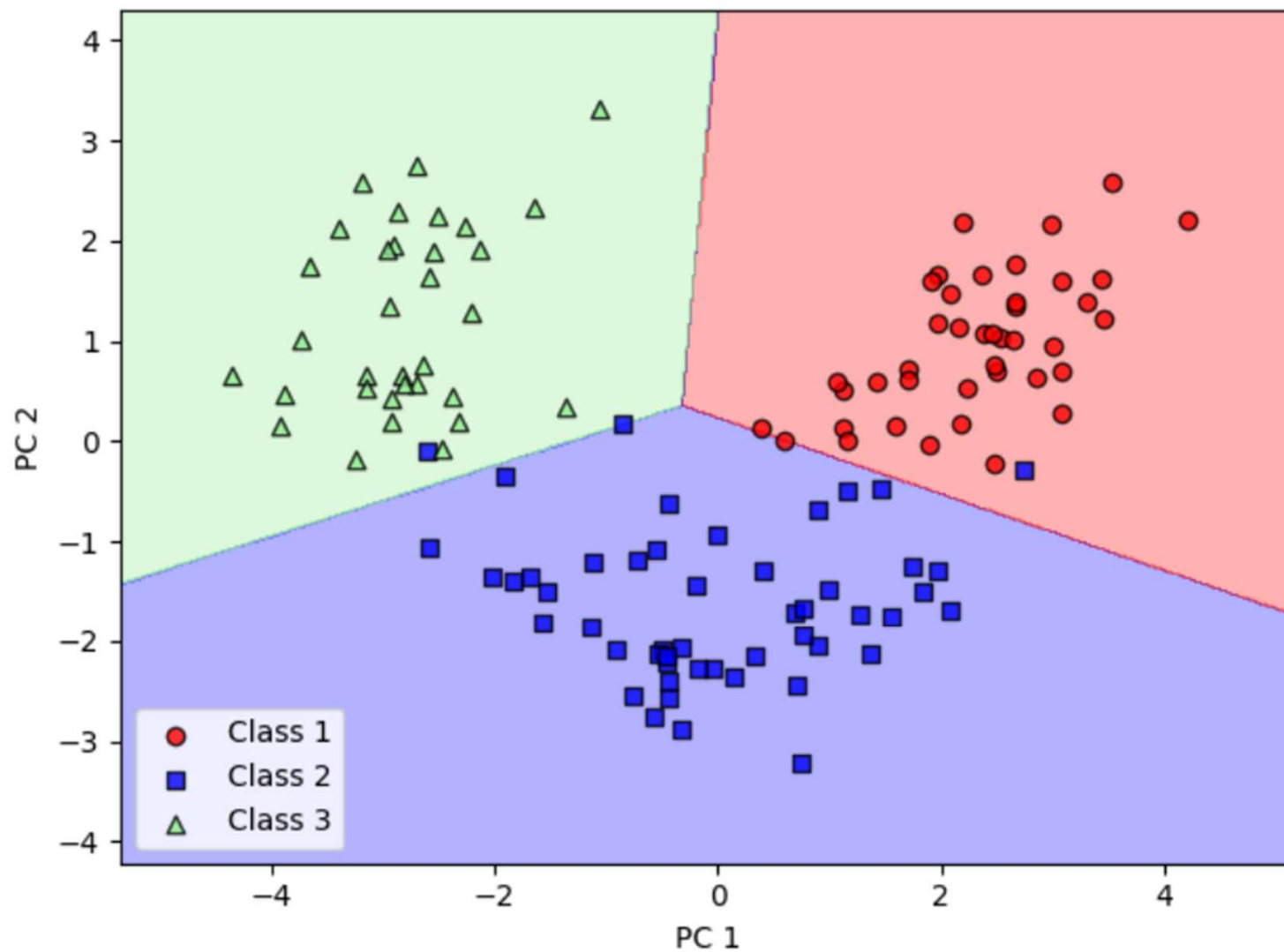


Transformed
wine
dataset



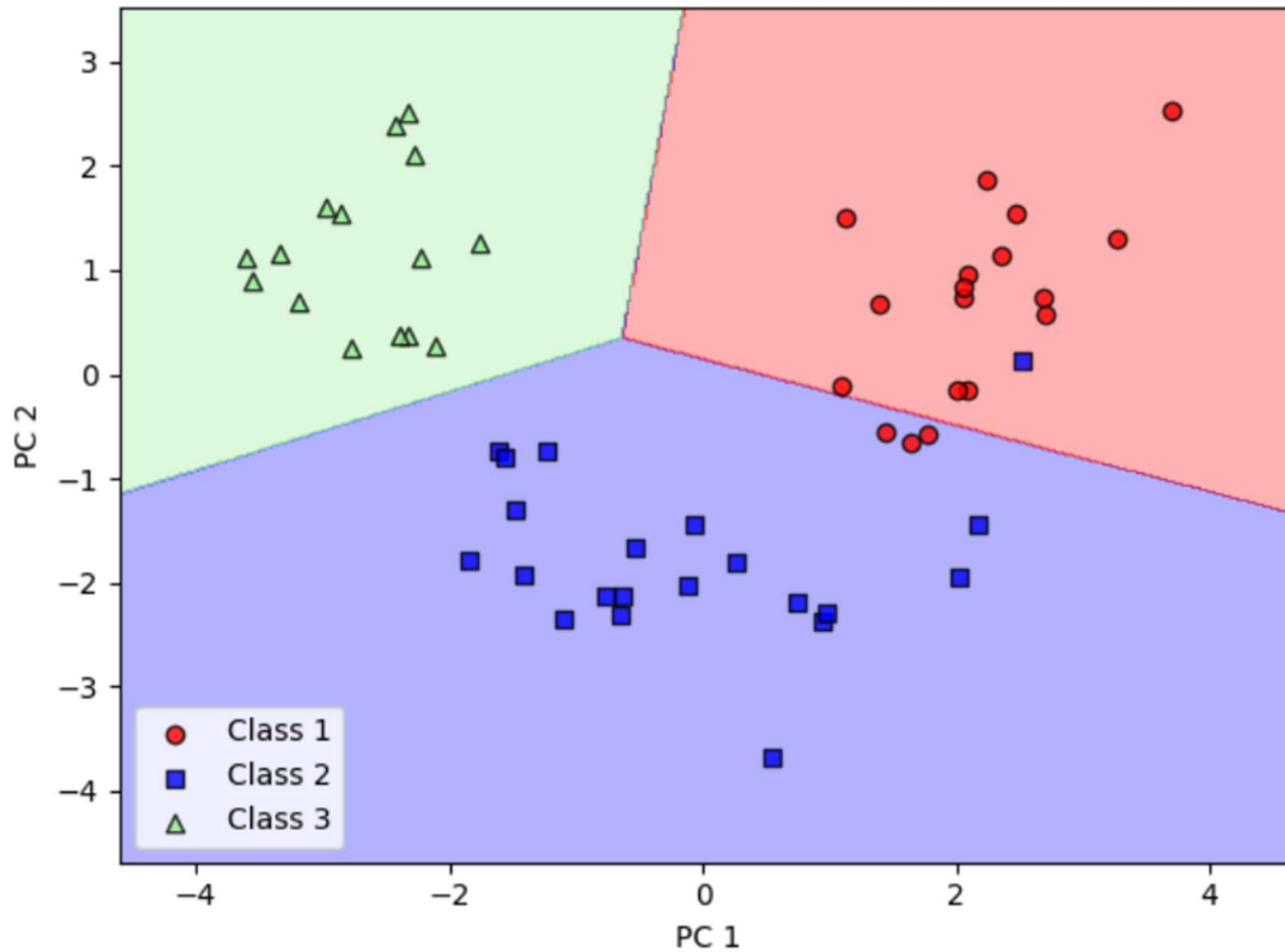
PCA in scikit-learn

(no class label information)



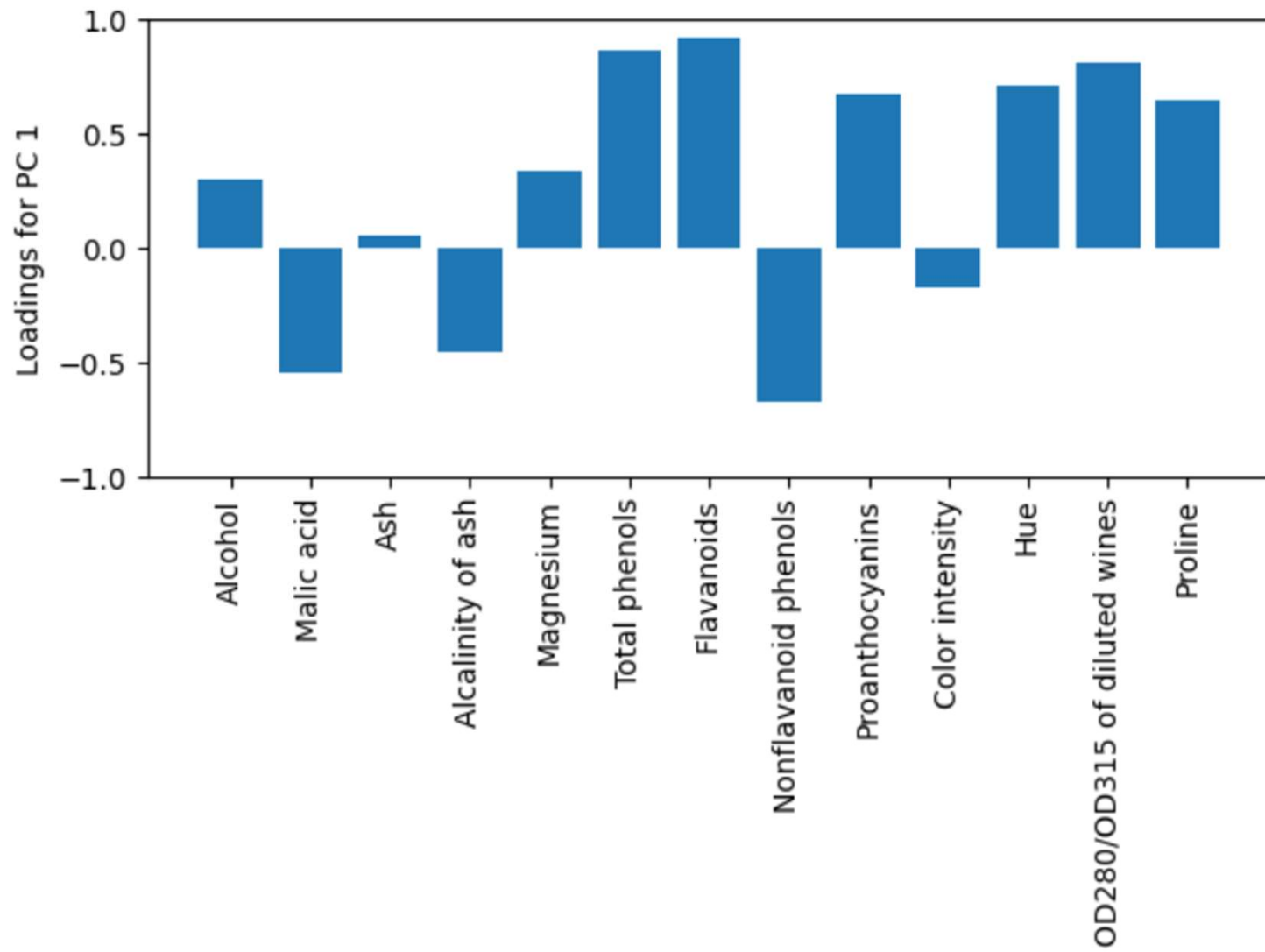
Logistic
regression
after PCA

(training examples)



Logistic
regression
after PCA

(test examples)

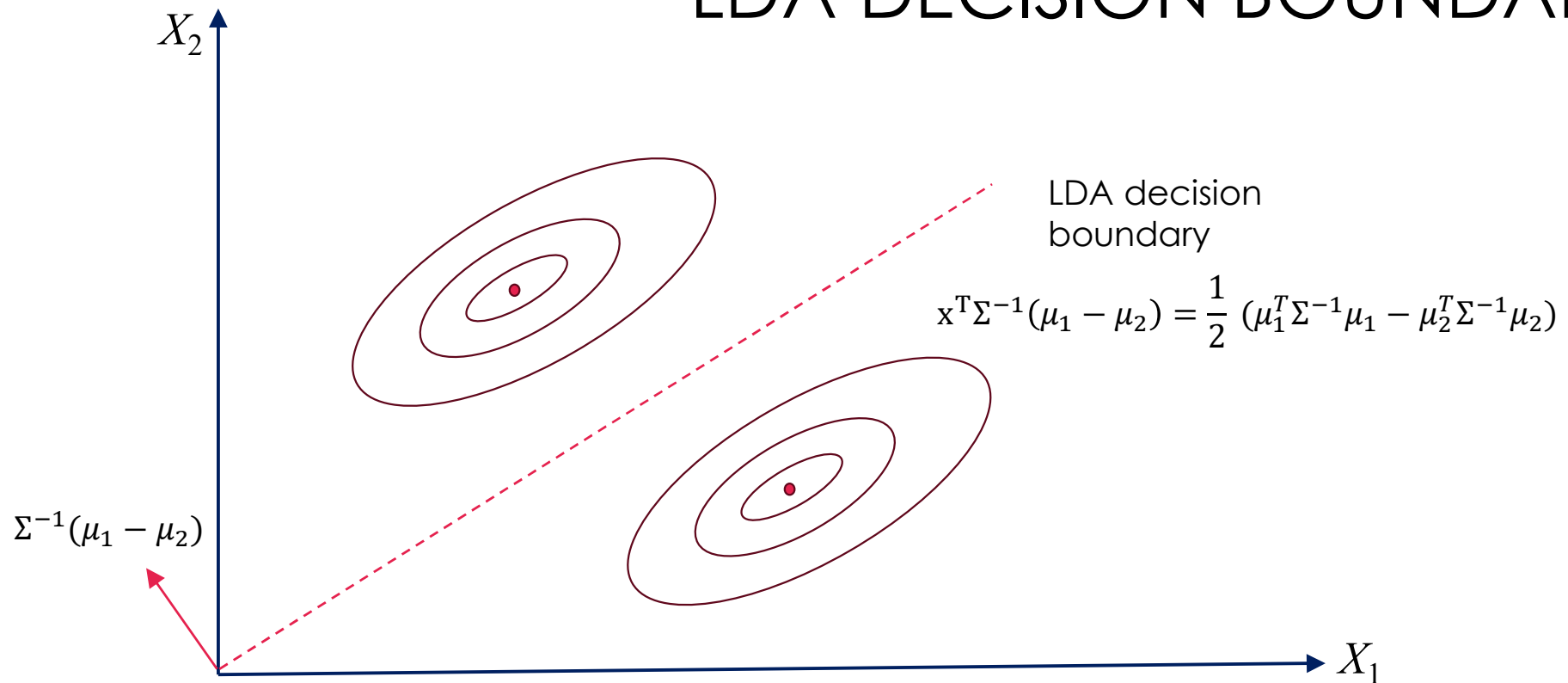


Assessing
feature
contributions

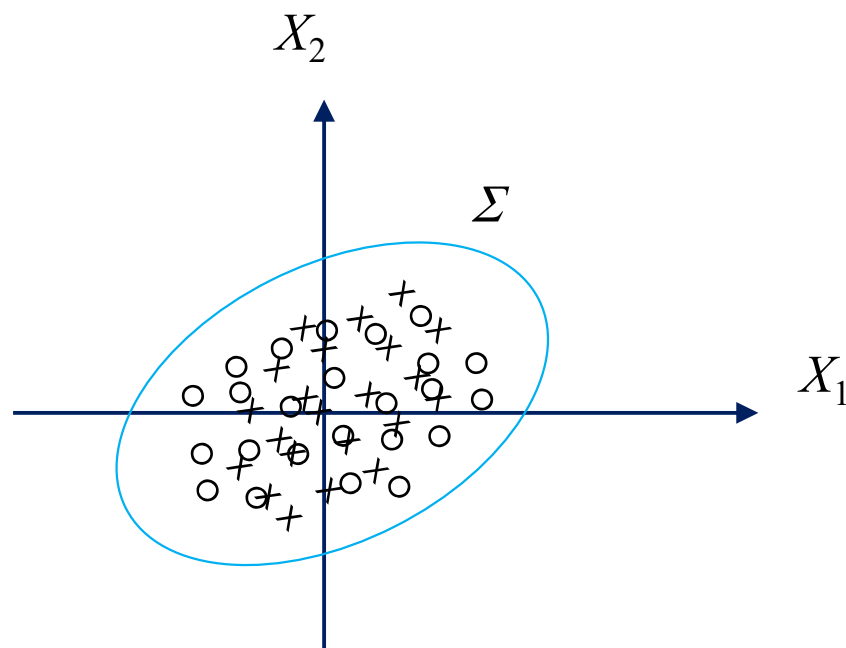
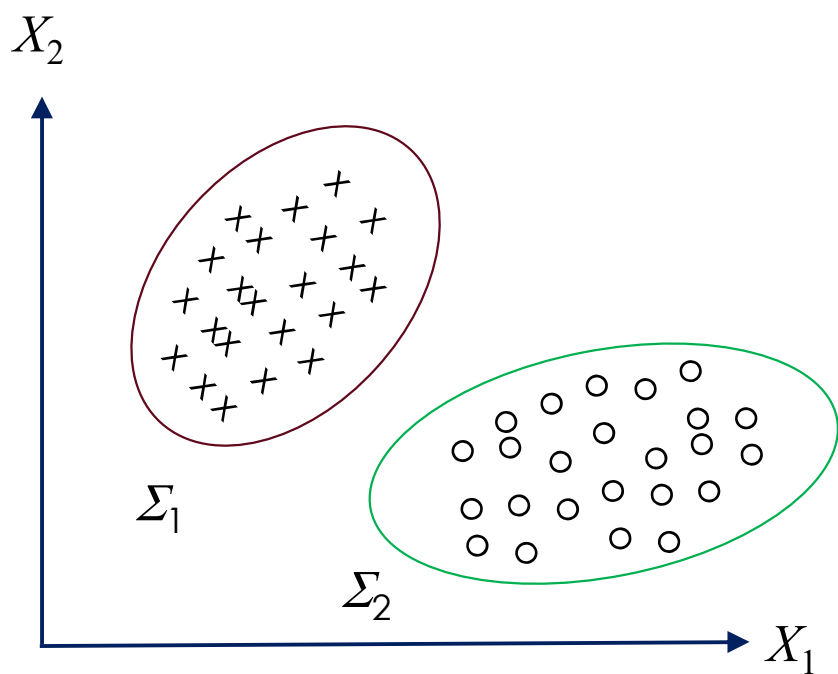


LINEAR DISCRIMINANT ANALYSIS (LDA)

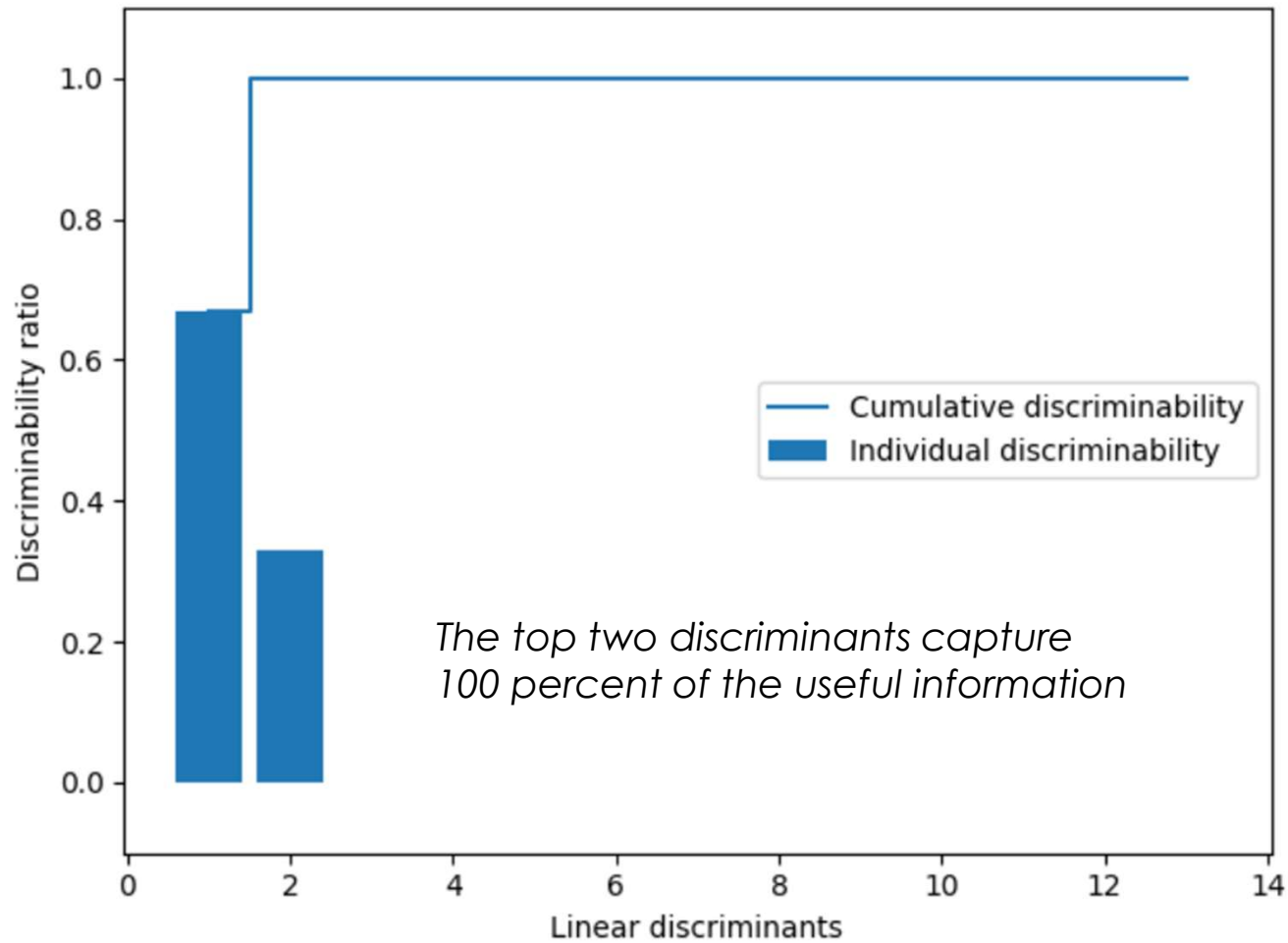
LDA DECISION BOUNDARY

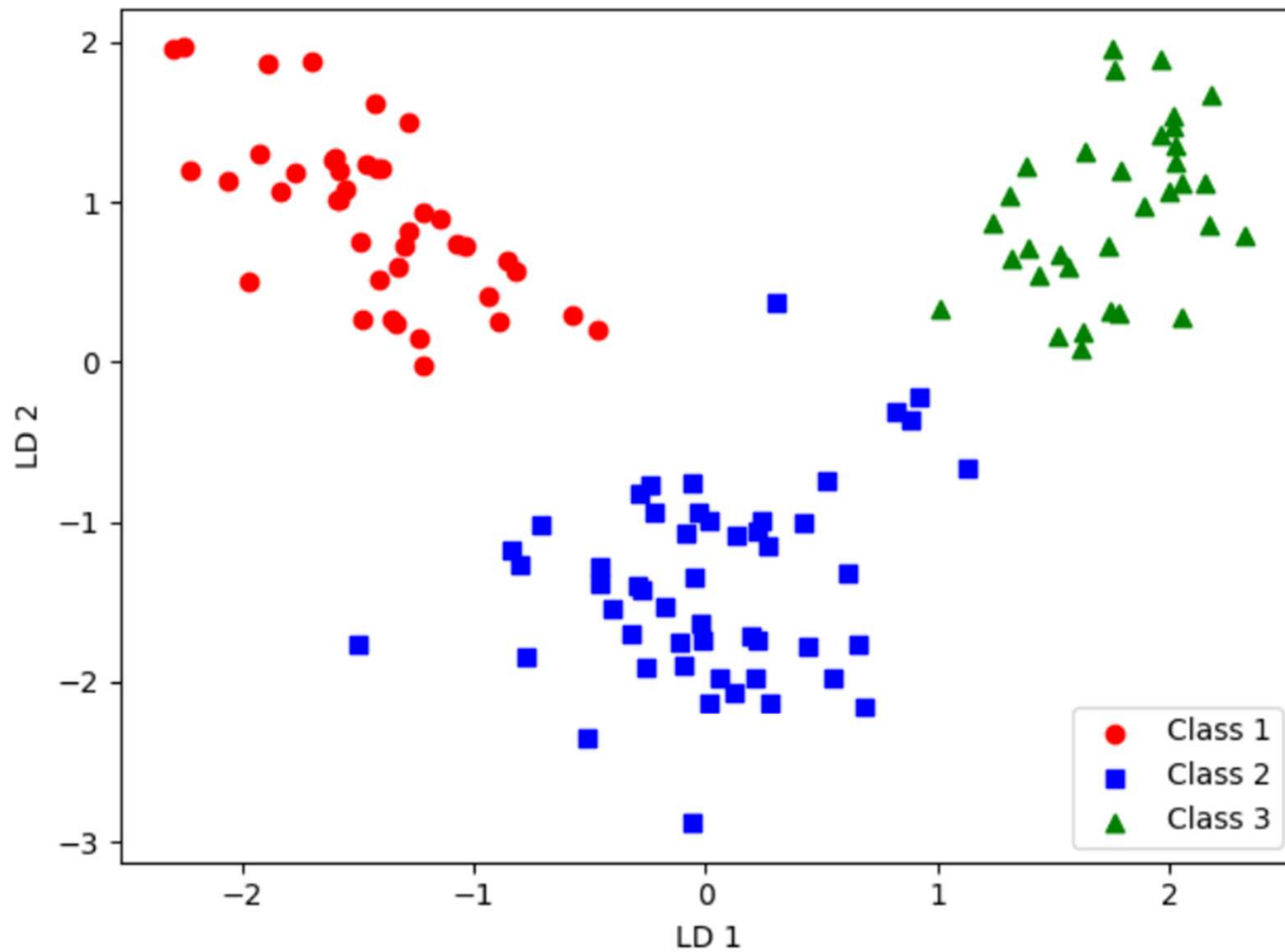


Pooled covariance



Discriminability

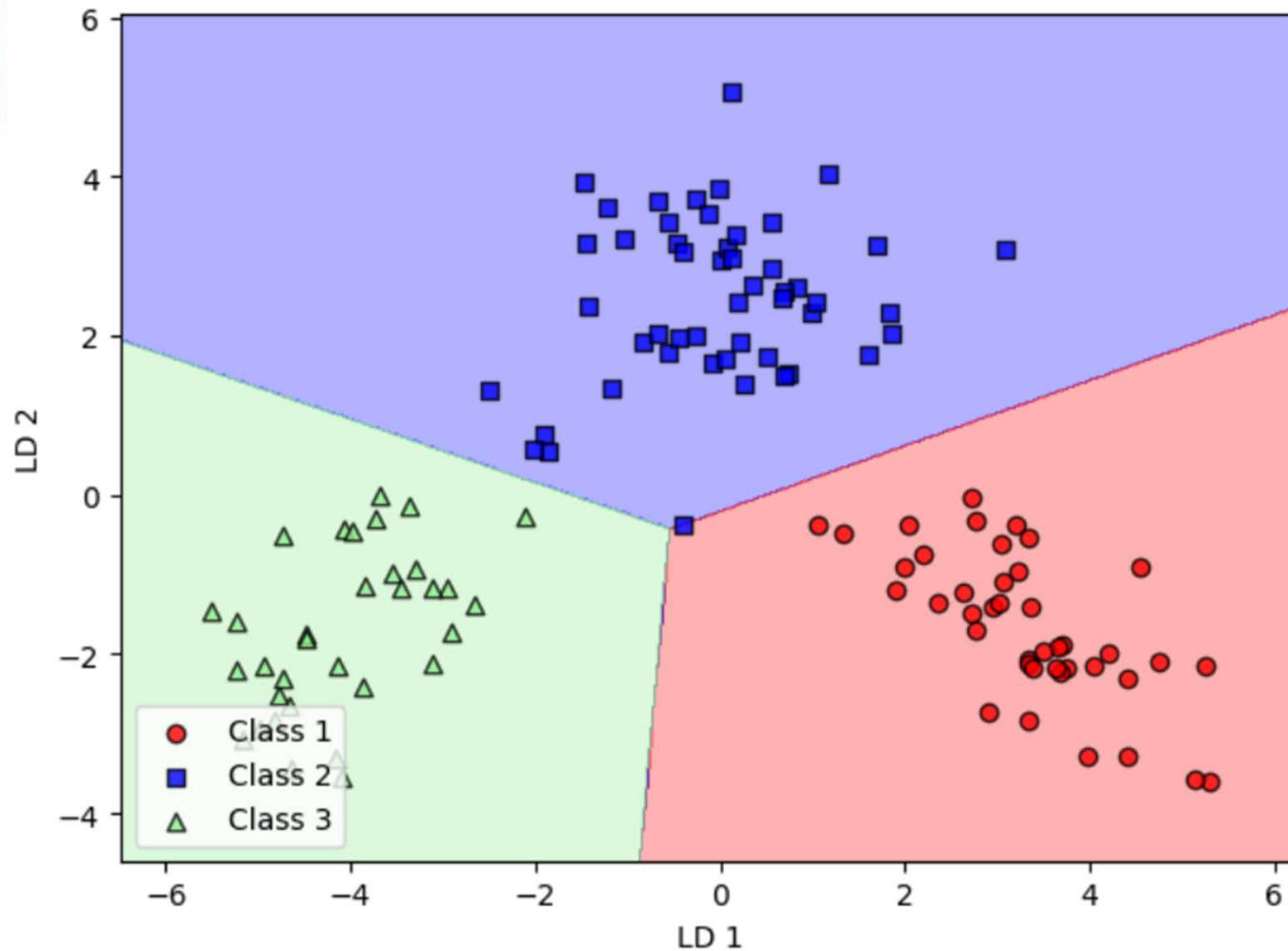




Projecting
wine data
onto first two
discriminants

LDA via scikit-learn

Logistic regression after
LDA transformation
(training examples)



Logistic regression after
LDA transformation
(test examples)





REFERENCES

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
- A. Ng. CS229 Lecture Notes. Stanford University. 2023.
- D. Kobak. Machine Learning I. University of Tübingen. 2021.